UNIT-I Introduction to Cloud Computing

- Overview
- Roots of Cloud Computing
- Layers and Types of Cloud
- Desired Features of a Cloud
- Benefits and Disadvantages of Cloud Computing
- Cloud Infrastructure Management
- Infrastructure as a Service Providers
- Platform as a Service Providers
- Challenges and Risks
- Assessing the role of Open Standards

CLOUD COMPUTING- Overview

"Cloud is a parallel and distributed computing system consisting of a collection of inter- connected and virtualised computers that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements (SLA) established through negotiation between the service provider and consumers."

The use of the word "cloud" makes reference to the two essential concepts:

- **Abstraction:** Cloud computing abstracts the details of system implementation from users and developers. Applications run on physical systems that aren't specified, data is stored in locations that are unknown, administration of systems is outsourced to others, and access by users is ubiquitous.
- **Virtualization:** Cloud computing virtualizes systems by pooling and sharing resources. Systems and storage can be provisioned as needed from a centralized infrastructure, costs are assessed on a metered basis, multi-tenancy is enabled, and resources are scalable with agility.

To help clarify how cloud computing has changed the nature of commercial system deployment, consider these three examples:

1. **Google:** In the last decade, Google has built a worldwide network of datacenters to service its search engine. In doing so Google has captured a substantial portion of the world's advertising revenue. That revenue has enabled Google to offer free software to users based on that infrastructure and has changed the market for user-facing software.
2. **Azure Platform:** By contrast, Microsoft is creating the Azure Platform. It enables.NET Framework applications to run over the Internet as an alternate platform for Microsoft developer software running on desktops.
3. **Amazon Web Services:** One of the most successful cloud-based businesses is Amazon Web

Services, which is an Infrastructure as a Service offering that lets you rent virtual computers on Amazon's own infrastructure.

These new capabilities enable applications to be written and deployed with minimal expense and to be rapidly scaled and made available worldwide as business conditions permit. This is truly a revolutionary change in the way enterprise computing is created and deployed.

<u>ROOTS OF CLOUD COMPUTING</u>

The roots of clouds computing by observing the advancement of several technologies, especially in

- Hardware (virtualization, multi-core chips),
- Internet technologies (Web services, service-oriented architectures, Web 2.0),
- Distributed computing (clusters, grids), and
- Systems management (autonomic computing, data center automation).

Following figure shows the convergence of technology fields that significantly advanced and contributed to the advent of cloud computing.

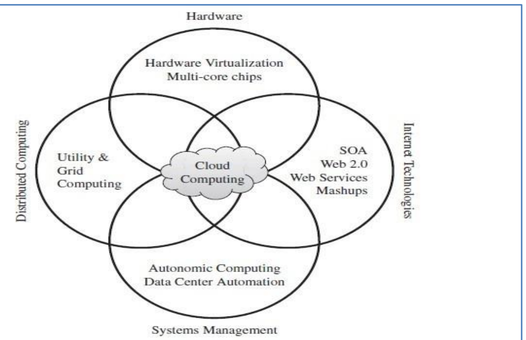

FIGURE 1.1. Convergence of various advances leading to the advent of cloud computing.

From Mainframes to Clouds

We are currently experiencing a switch in the IT, from in-house generated computing power into utility-supplied computing resources delivered over the Internet as Web services.

Computing delivered as a utility can be defined as “on demand delivery of infrastructure, applications, and business processes in a security-rich, shared, scalable, and based computer environment over the Internet for a fee”. This model brings benefits to both consumers and providers of IT services.

Consumers can attain reduction on IT-related costs by choosing to obtain cheaper services from external providers as opposed to heavily investing on IT infrastructure and personnel hiring. The “on-demand” component of this model allows consumers to adapt their IT usage to rapidly increasing or unpredictable computing needs.

Providers of IT services achieve better operational costs; hardware and software infrastructures are built to provide multiple solutions and serve many users, thus increasing efficiency and ultimately leading to faster return on investment (ROI) as well as lower total cost of ownership (TCO).

Grid Computing

Grid computing enables aggregation of distributed resources and transparently access to them. Most production grids seek to share compute and storage resources distributed across different administrative domains. Its main focus is being speeding up a broad range of scientific applications, such as climate modeling, drug design, and protein analysis.

A key aspect of the grid vision realization has been building standard Web services-based protocols that allow distributed resources to be "discovered, accessed, allocated, monitored, accounted for, and billed for, etc., and in general managed as a single virtual system."

Utility Computing

In utility computing environments, users assign a "utility" value to their jobs, where utility is a fixed or time-varying valuation that captures various QoS constraints (deadline, importance, satisfaction). The valuation is the amount they are willing to pay a service provider to satisfy their demands. The service providers then attempt to maximize their own utility, where said utility may directly correlate with their profit.

Hardware Virtualization

The idea of virtualizing a computer system's resources, including processors, memory, and I/O devices, has been well established for decades, aiming at improving sharing and utilization of computer systems. Hardware virtualization allows running multiple operating systems and software stacks on a single physical platform. The virtual machine monitor (VMM), also called a hypervisor, mediates access to the physical hardware presenting to each guest operating system a virtual machine (VM), which is a set of virtual platform interfaces.

The advent of several innovative technologies—multi-core chips, paravirtualization, hardware-assisted virtualization, and live migration of VMs—has contributed to an increasing adoption of virtualization on server systems.

Traditionally, perceived benefits were improvements on sharing and utilization, better manageability, and higher reliability.

KVM. The kernel-based virtual machine (KVM) is a Linux virtualization subsystem. Is has been part of the mainline Linux kernel since version 2.6.20, thus being natively supported by several distributions. In addition, activities such as memory management and scheduling are carried out by existing kernel features, thus making KVM simpler and smaller than hypervisors that take control of the entire machine.

KVM leverages hardware-assisted virtualization, which improves performance and allows it to support unmodified guest operating systems ; currently, it supports several versions of Windows, Linux, and UNIX.

Virtual Appliances and the Open Virtualization Format

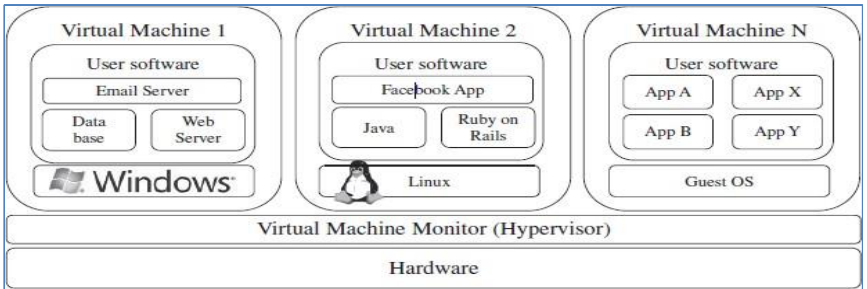

FIGURE 1.2. A hardware virtualized server hosting three virtual machines, each one running distinct operating system and user level software stack.

An application combined with the environment needed to run it (operating system, libraries, compilers, databases, application containers, and so forth) is referred to as a “virtual appliance.” Packaging application environments in the shape of virtual appliances eases software customization, configuration, and patching and improves portability. Most commonly, an appliance is shaped as a VM disk image associated with hardware requirements, and it can be readily deployed in a hypervisor.

In a multitude of hypervisors, where each one supports a different VM image format and the formats are incompatible with one another, a great deal of interoperability issues arises. For instance, Amazon has its Amazon machine image (AMI) format, made popular on the Amazon EC2 public cloud. Other formats are used by Citrix XenServer, several Linux distributions that ship with KVM, Microsoft Hyper-V, and VMware ESX.

Autonomic Computing

Autonomic, or self-managing, systems rely on monitoring probes and gauges (sensors), on an adaptation engine (autonomic manager) for computing optimizations based on monitoring data, and on effectors to carry out changes on the system. IBM’s autonomic Computing Initiative has contributed to define the four properties of autonomic systems: self-configuration, self-optimization, self-healing, and self-protection. IBM has also suggested a reference model for autonomic control loops of autonomic managers, called MAPE-K (Monitor Analyze Plan Execute—Knowledge).

LAYERS AND TYPES OF CLOUDS

Cloud computing services are divided into three classes, according to the abstraction level of the capability provided and the service model of providers, namely:

(1) Infrastructure as a Service,

(2) Platform as a Service, and

(3) Software as a Service.

Figure 1.3 depicts the layered organization of the cloud stack from physical infrastructure to applications. These abstraction levels can also be viewed as a layered architecture where services of a higher layer can be composed from services of the underlying layer.

Cloud development environments are built on top of infrastructure services to offer application development and deployment capabilities; in this level, various programming models, libraries, APIs, and mashup editors enable the creation of a range of business, Web, and scientific applications. Once deployed in the cloud, these applications can be consumed by end users.

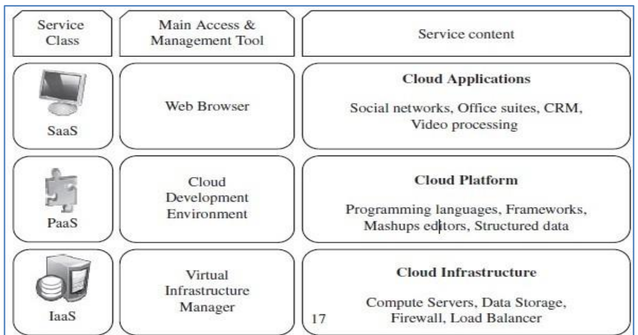

FIGURE 1.3. The cloud computing stack.

Infrastructure as a Service (IaaS)

Offering virtualized resources (computation, storage, and communication) on demand is known as Infrastructure as a Service (IaaS). A cloud infrastructure enables on-demand provisioning of servers running several choices of operating systems and a customized software stack. Infrastructure services are considered to be the bottom layer of cloud computing systems.

Amazon Web Services mainly offers IaaS, which in the case of its EC2 service means offering VMs with a software stack that can be customized similar to how an ordinary physical server would be customized. Users are given privileges to perform numerous activities to the server, such as: starting and stopping it, customizing it by installing software packages, attaching virtual disks to it, and configuring access permissions and firewalls rules.

Platform as a Service (PaaS)

In addition to infrastructure-oriented clouds that provide raw computing and storage services, another approach is to offer a higher level of abstraction to make a cloud easily programmable, known as Platform as a Service (PaaS). A cloud platform offers an environment on which developers create and deploy applications and do not necessarily need to know how many processors or how much memory that applications will be using. In addition, multiple programming models and specialized services (e.g., data access, authentication, and payments) are offered as building blocks to new applications. Google AppEngine, an example of Platform as a Service, offers a scalable environment for developing and hosting Web applications, which should be written in specific programming languages such as Python or Java, and use the services' own proprietary structured object data store. Building blocks include an in-memory object cache (memcache), mail service, instant messaging service (XMPP), an image manipulation service, and integration with Google Accounts authentication service.

Software as a Service (SaaS)

Applications reside on the top of the cloud stack. Services provided by this layer can be accessed by end users through Web portals. Therefore, consumers are increasingly shifting from locally installed computer programs to on-line software services that offer the same functionally. Traditional desktop applications such as word processing and spreadsheet can now be accessed as a service in the Web. This model of delivering applications, known as Software as a Service (SaaS), alleviates the burden of software maintenance for customers and simplifies development and testing for providers. Salesforce.com, which relies on the SaaS model, offers business productivity applications (CRM) that reside completely on their servers, allowing costumers to customize and access applications on demand.

Deployment Models

Although cloud computing has emerged mainly from the appearance of public computing utilities, other deployment models, with variations in physical location and distribution, have been adopted. In this sense, regardless of its service class, a cloud can be classified as –

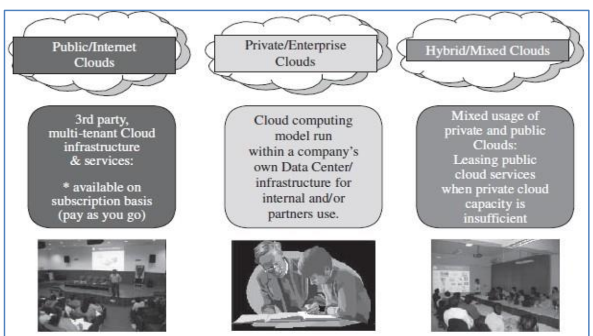

FIGURE 1.4. Types of clouds based on deployment models.

A) Public Cloud
B) Private Cloud
C) Community Cloud
D) Hybrid Cloud

A) **Public cloud -** “Cloud made available in a pay-as-you-go manner to the general public”.
B) **Private cloud** – “Internal data center of a business or other organization, not made available to the general public.”
C) **Community cloud -** “It is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations).”
D) **Hybrid cloud –** It takes shape when a private cloud is supplemented with computing capacity from public clouds. The approach of temporarily renting capacity to handle spikes in load is known as “cloud-bursting”.

DESIRED FEATURES OF A CLOUD:

Certain features of a cloud are essential to enable services that truly represent the cloud computing model and satisfy expectations of consumers. They are as follows -

1) **On-demand self-service:** A client can provision computer resources without the need for interaction with cloud service provider personnel.
2) **Broad network access:** Access to resources in the cloud is available over the network using standard methods in a manner that provides platform-independent access to clients of all types. This includes a mixture of heterogeneous operating systems, and thick and thin platforms such as laptops, mobile phones, and PDA.
3) **Resource pooling:** Pooling is the idea of abstraction that hides the location of resources such as virtual machines, processing, memory, storage, and network bandwidth and connectivity. A cloud service provider creates resources that are pooled together in a system that supports multi-tenant usage. Physical and virtual systems are dynamically allocated or reallocated as needed.
4) **Rapid elasticity:** Resources can be rapidly and elastically provisioned. The system can add resources by either scaling up systems (more powerful computers) or scaling out systems (more computers of the same kind), and scaling may be automatic or manual.
5) **Measured service:** The use of cloud system resources is measured, audited, and reported to the customer based on a metered system. A client can be charged based on a known metric such as amount of storage used, number of transactions, network I/O (Input/Output) or bandwidth, amount of processing power used, and so forth. A client is charged based on the level of services provided.

BENEFITS OF CLOUD COMPUTING

1) **Lower costs:** Because cloud networks operate at higher efficiencies and with greater utilization, significant cost reductions are often encountered.
2) **Ease of utilization:** Depending upon the type of service being offered, you may find that you do not

require hardware or software licenses to implement your service.

3) **Quality of Service:** The Quality of Service (QoS) is something that you can obtain under contract from your vendor.
4) **Reliability:** The scale of cloud computing networks and their ability to provide load balancing and failover makes them highly reliable, often much more reliable than what you can achieve in a single organization.
5) **Outsourced IT management:** A cloud computing deployment lets someone else manage your computing infrastructure while you manage your business. In most instances, you achieve considerable reductions in IT staffing costs.
6) **Simplified maintenance and upgrade:** Because the system is centralized, you can easily apply patches and upgrades. This means your users always have access to the latest software versions.
7) **Low Barrier to Entry:** In particular, upfront capital expenditures are dramatically reduced. In cloud computing, anyone can be a giant at any time.

DISADVANTAGES OF CLOUD COMPUTING

While the benefits of cloud computing are myriad, the disadvantages are just as numerous.

1) As a general rule, the advantages of cloud computing present a more compelling case for small organizations than for larger ones. Larger organizations can support IT staff and development efforts that put in place custom software solutions that are crafted with their particular needs in mind.
2) When you use an application or service in the cloud, you are using something that isn't necessarily as customizable as you might want. Additionally, although many cloud computing applications are very capable, applications deployed on-premises still have many more features than their cloud counterparts.
3) All cloud computing applications suffer from the inherent latency that is intrinsic in their WAN connectivity. While cloud computing applications excel at large-scale processing tasks, if your application needs large amounts of data transfer, cloud computing may not be the best model for you.
4) In order for communication to survive on a distributed system, it is necessarily unidirectional in nature. All the requests you use in HTTP: PUTs, GETs, and so on are requests to a service provider. The service provider then sends a response.
5) Although it may seem that you are carrying on a conversation between client and provider, there is an architectural disconnect between the two. That lack of state allows messages to travel over different routes and for data to arrive out of sequence, and many other characteristics allow the communication to succeed even when the medium is faulty. Therefore, to impose transactional coherency upon the system, additional overhead in the form of service brokers, transaction managers, and other middleware

must be added to the system. This can introduce a very large performance hit into some applications.

CLOUD INFRASTRUCTURE MANAGEMENT:

A key challenge IaaS providers face when building a cloud infrastructure is managing physical and virtual resources, namely servers, storage, and networks, in a holistic fashion. The orchestration of resources must be performed in a way to rapidly and dynamically provision resources to applications.

The software toolkit responsible for this orchestration is called a virtual infrastructure manager (VIM). This type of software resembles a traditional operating system—but instead of dealing with a single computer, it aggregates resources from multiple computers, presenting a uniform view to user and applications. The term "cloud operating system" is also used to refer to it. Other terms include "Infrastructure Sharing Software" and "Virtual Infrastructure Engine."

The availability of a remote cloud-like interface and the ability of managing many users and their permissions are the primary features. It distinguishes the "Cloud Toolkits" from "VIMs".

Features: We now present a list of both basic and advanced features that are usually available in VIMs.

a) **Virtualization Support:** The multi-tenancy aspect of clouds requires multiple customers with disparate requirements to be served by a single hardware infrastructure. Virtualized resources (CPUs, memory, etc.) can be sized and resized with certain flexibility. These features make hardware virtualization, the ideal technology to create a virtual infrastructure that partitions a data center among multiple tenants.

b) **Self-Service, On-Demand Resource Provisioning:** Self-service access to resources has been supposed as one the most attractive features of clouds. This feature enables users to directly obtain services from clouds, without interacting with a human system administrator. This capability "eliminates the need for more time-consuming, labor-intensive, human driven procurement processes familiar to many in IT". Therefore, exposing a self-service interface, through which users can easily interact with the system, is a highly desirable feature of a Virtualized Infrastructure Manager (VIM).

c) **Multiple Backend Hypervisors:** Different virtualization models and tools offer different benefits, drawbacks, and limitations. Thus, some VI managers provide a uniform management layer regardless of the virtualization technology used. This characteristic is more visible in open-source VI managers, which usually provide pluggable drivers to interact with multiple hypervisors. In this direction, the aim of libvirt is to provide a uniform API that VI managers can use to manage domains in virtualized nodes.

d) **Storage Virtualization:** Virtualizing storage means abstracting logical storage from physical storage. By consolidating all available storage devices in a data center, it allows creating virtual disks independent from device and location. Storage devices are commonly organized in a storage area network (SAN) and attached to servers via protocols such as Fibre Channel, iSCSI, and NFS; a storage

Commented [MAPG1]: All-inclusive

Commented [MAPG2]: different

Commented [MAPG3]: Also known as a Virtual Machine Monitor(VMM), is software that creates and runs virtual machines (VMs)

Commented [MAPG4]: libvirt is an open-source API.

It is a toolkit to manage virtualization platforms controller provides the layer of abstraction between virtual and physical storage.

e) Interface to Public Clouds: Researchers have perceived that extending the capacity of a local in-house computing infrastructure by borrowing resources from public clouds is advantageous. In this fashion, institutions can make good use of their available resources and, in case of spikes in demand, extra load can be offloaded to rented resources.

f) Virtual Networking: Virtual networks allow creating an isolated network on top of a physical infrastructure independently from physical topology and locations. A virtual LAN (VLAN) allows isolating traffic that shares a switched network, allowing VMs to be grouped into the same broadcast domain.

Support for creating and configuring virtual networks to group VMs placed throughout a data center is provided by most VI managers. The VI managers that interface with public clouds often support secure VPNs connecting local and remote VMs.

g) Dynamic Resource Allocation: Increased awareness of energy consumption in data centers has encouraged the practice of dynamic consolidating VMs in a fewer number of servers. In cloud infrastructures, where applications have variable and dynamic needs, capacity management and demand prediction are especially complicated. This fact triggers the need for dynamic resource allocation aiming at obtaining a timely match of supply and demand.

Energy consumption reduction and better management of SLAs can be achieved by dynamically remapping VMs to physical machines at regular intervals. Machines that are not assigned any VM can be turned off or put on a low power state. In the same fashion, overheating can be avoided by moving load away from hotspots.

A number of VI managers include a dynamic resource allocation feature that continuously monitors utilization across resource pools and reallocates available resources among VMs according to application needs.

h) Reservation and Negotiation Mechanism: When users request computational resources to available at a specific time, requests are termed advance reservations (AR), in contrast to best-effort requests, when users request resources whenever available. To support complex requests, such as AR, a VI manager must allow users to "lease" resources expressing more complex terms. This is especially useful in clouds on which resources are scarce; since not all requests may be satisfied immediately, they can benefit of VM placement strategies that support queues, priorities, and advance reservations.

Additionally, leases may be negotiated and renegotiated, allowing provider and consumer to modify a lease or present counter proposals until an agreement is reached. This feature is illustrated by the case in which an AR request for a given slot cannot be satisfied, but the provider can offer a distinct slot that is still satisfactory to the user.

i) High Availability and Data Recovery: The high availability (HA) feature of VI managers aims at minimizing application downtime and preventing business disruption. A few VI managers accomplish this by providing a failover mechanism, which detects failure of both physical and virtual servers and restarts VMs on healthy physical servers. This style of HA protects from host, but not VM, failures.

In this sense, some VI managers offer data protection mechanisms that perform incremental backups of VM images.

INFRASTRUCTURE AS A SERVICE PROVIDERS

Public Infrastructure as a Service providers commonly offer

- virtual servers containing one or more CPUs,
- running several choices of operating systems and
- a customized software stack.
- In addition, storage space and communication facilities are often provided.

Features

In spite of being based on a common set of features, IaaS offerings can be distinguished by the availability of specialized features that influence the cost-benefit ratio to be experienced by user applications when moved to the cloud. The most relevant features are:

(i) Geographic distribution of data centers;
(ii) Variety of user interfaces and APIs to access the system
(iii) Specialized components and services that aid particular applications (e.g., load balancers, firewalls)
(iv) Choice of virtualization platform and operating systems; and
(v) Different billing methods and period (e.g., prepaid vs. post-paid, hourly vs. monthly).

(i) Geographic Presence: To improve availability and responsiveness, a provider of worldwide services would typically build several data centers distributed around the world. For example, Amazon Web Services(AWS) presents the concept of "availability zones" and "regions" for its EC2 (Elastic Compute Cloud) service. Availability zones are "distinct locations that are engineered to be insulated from failures in other availability zones and provide inexpensive, low-latency network connectivity to other availability zones in the same region." Regions, in turn, "are geographically dispersed and will be in separate geographic areas or countries."

(ii) **User Interfaces and Access to Servers:** Ideally, a public IaaS provider must provide multiple access means to its cloud, thus catering for various users and their preferences. Different types of user interfaces (UI) provide different levels of abstraction, the most common being graphical user interfaces (GUI), command-line tools (CLI), and Web service (WS) APIs.

GUIs are preferred by end users who need to launch, customize, and monitor a few virtual servers and do not necessary need to repeat the process several times. On the other hand, CLIs offer more flexibility and the possibility of automating repetitive tasks via scripts (e.g., start and shutdown a number of virtual servers at regular intervals). WS APIs offer programmatic access to a cloud using standard HTTP requests, thus allowing complex services to be built on top of IaaS clouds.

Advance Reservation of Capacity

Advance reservations allow users to request for an IaaS provider to reserve resources for a specific time frame in the future, thus ensuring that cloud resources will be available at that time. However, most clouds only support best-effort requests; that is, users requests are server whenever resources are available.

Amazon Reserved Instances is a form of advance reservation of capacity, allowing users to pay a fixed amount of money in advance to guarantee resource availability at anytime during an agreed period and then paying a discounted hourly rate when resources are in use. However, only long periods of 1 to 3 years are offered; therefore, users cannot express their reservations in finer granularities—for example, hours or days.

Automatic Scaling and Load Balancing

Automatic scaling is a highly desirable feature of IaaS clouds. It allow users to set conditions for when they want their applications to scale up and down, based on application- specific metrics such as transactions per second, number of simultaneous users, request latency, and so forth. When the number of virtual servers is increased by automatic scaling, incoming traffic must be automatically distributed among the available servers. This activity enables applications to promptly respond to traffic increase while also achieving greater fault tolerance.

Service-Level Agreement

Service-level agreements (SLAs) are offered by IaaS providers to express their commitment to delivery of a certain QoS. To customers it serves as a warranty. An SLA usually include availability and performance guarantees. Additionally, metrics must be agreed upon by all parties as well as penalties for violating these expectations.

Most IaaS providers focus their SLA terms on availability guarantees, specifying the minimum percentage of time the system will be available during a certain period. For instance, Amazon EC2 states that “if the annual uptime Percentage for a customer drops below 99.95% for the service year, that customer is eligible to receive a service credit equal to 10% of their bill.”

Hypervisor and Operating System Choice

Traditionally, IaaS offerings have been based on heavily customized open-source Xen deployments. IaaS providers needed expertise in Linux, networking, virtualization, metering, resource management, and many other low-level aspects to successfully deploy and maintain their cloud offerings. More recently, there has been an emergence of turnkey IaaS platforms such as VMWare vCloud and Citrix Cloud Center (C3) which have lowered the barrier of entry for IaaS competitors, leading to a rapid expansion in the IaaS marketplace.

PLATFORM AS A SERVICE PROVIDERS

Public Platform as a Service providers commonly offer a development and deployment environment that allow users to create and run their applications with little or no concern to low- level details of the platform. In addition, specific programming languages and frameworks are made available in the platform, as well as other services such as persistent data storage and in-memory caches.

Features

Programming Models, Languages, and Frameworks

Programming models made available by IaaS providers define how users can express their applications using higher levels of abstraction and efficiently run them on the cloud platform. Each model aims at efficiently solving a particular problem. In the cloud computing domain, the most common activities that require specialized models are: processing of large dataset in clusters of computers (MapReduce model), development of request-based Web services and applications; definition and orchestration of business processes in the form of workflows (Workflow model); and high-performance distributed execution of various computational tasks.

For user convenience, PaaS providers usually support multiple programming languages. Most commonly used languages in platforms include Python and Java (e.g., Google AppEngine), .NET languages (e.g., Microsoft Azure), and Ruby (e.g., Heroku). Force.com has devised its own programming language (Apex) and an Excel-like query language, which provide higher levels of abstraction to key platform functionalities.

A variety of software frameworks are usually made available to PaaS developers, depending on application focus. Providers that focus on Web and enterprise application hosting offer popular frameworks such as Ruby on Rails, Spring, Java EE, and .NET.

Persistence Options

A persistence layer is essential to allow applications to record their state and recover it in case of crashes, as well as to store user data.

CHALLENGES AND RISKS

Despite the initial success and popularity of the cloud computing paradigm and the extensive availability of providers and tools, a significant number of challenges and risks are inherent to this new model of computing. Providers, developers, and end users must consider these challenges and risks to take good advantage of cloud computing. Issues to be faced include user privacy, data security, data lock-in, availability of service, disaster recovery, performance, scalability, energy-efficiency, and programmability.

a) Security, Privacy, and Trust: Ambrust et al. cite information security as a main issue: "current cloud offerings are essentially public which exposes the system to more attacks." For this reason there are potentially additional challenges to make cloud computing environments as secure as in-house IT systems. At the same time, existing, well understood technologies can be leveraged, such as data encryption, VLANs, and firewalls.

Security and privacy affect the entire cloud computing stack, since there is a massive use of third-party services and infrastructures that are used to host important data or to perform critical operations. In this scenario, the trust toward providers is fundamental to ensure the desired level of privacy for applications hosted in the cloud.

Legal and regulatory issues also need attention. When data are moved into the Cloud, providers may choose to locate them anywhere on the planet. The physical location of data centers determines the set of laws that can be applied to the management of data. For example, specific cryptography techniques could not be used because they are not allowed in some countries. Similarly, country laws can impose that sensitive data, such as patient health records, are to be stored within national borders.

b) Data Lock-In and Standardization: A major concern of cloud computing users is about having their data locked-in by a certain provider. Users may want to move data and applications out from a provider that does not meet their requirements. However, in their current form, cloud computing infrastructures and platforms do not employ standard methods of storing user data and applications. Consequently, they do not interoperate and user data are not portable.

The Cloud Computing Interoperability Forum (CCIF) was formed by organizations such as Intel, Sun, and Cisco in order to "enable a global cloud computing ecosystem whereby organizations are able to seamlessly work together for the purposes for wider industry adoption of cloud computing technology." The development of the Unified Cloud Interface (UCI) by CCIF aims at creating a standard programmatic point of access to an entire cloud infrastructure. In the hardware virtualization sphere, the Open Virtual Format (OVF) aims at facilitating packing and distribution of software to be run on VMs so that virtual appliances can be made portable—that is, seamlessly run on hypervisor of different vendors.

c) Availability, Fault-Tolerance, and Disaster Recovery: It is expected that users will have certain expectations about the service level to be provided once their applications are moved to the cloud. These expectations include availability of the service, its overall performance, and what measures are to be taken when something goes wrong in the system or its components.

SLAs, which include QoS requirements, must be ideally set up between customers and cloud computing providers to act as warranty. An SLA specifies the details of the service to be provided, including availability and performance guarantees. Additionally, metrics must be agreed upon by all parties, and penalties for violating the expectations must also be approved.

d) Resource Management and Energy-Efficiency: One important challenge faced by providers of cloud computing services is the efficient management of virtualized resource pools. Physical resources such as CPU cores, disk space, and network bandwidth must be sliced and shared among virtual machines running potentially heterogeneous workloads. The multi-dimensional nature of virtual machines complicates the activity of finding a good mapping of VMs onto available physical hosts while maximizing user utility. Dimensions to be considered include: number of CPUs, amount of memory, size of virtual disks, and network bandwidth. Dynamic VM mapping policies may leverage the ability to suspend, migrate, and resume VMs as an easy way of preempting low-priority allocations in favor of higher-priority ones. Migration of VMs also brings additional challenges such as detecting when to initiate a migration, which VM to migrate, and where to migrate. In addition, policies may take advantage of live migration of virtual machines to relocate data center load without significantly disrupting running services. In this case, an additional concern is the trade- off between the negative impact of a live migration on the performance and stability of a service and the benefits to be achieved with that migration.

Another challenge concerns the outstanding amount of data to be managed in various VM management activities. Such data amount is a result of particular abilities of virtual machines, including the ability of traveling through space (i.e., migration) and time (i.e., checkpointing and rewinding) , operations that may be required in load balancing, backup, and recovery scenarios. In addition, dynamic provisioning of new VMs and replicating existing VMs require efficient mechanisms to make VM block storage devices (e.g., image files) quickly available at selected hosts.

* * * * *